# Additional deployment options

## Cloud Manager 3.7

NetApp
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# Table of Contents

Additional deployment options . . . 1
- Cloud Manager host requirements . . . 1
- Installing Cloud Manager on an existing Linux host . . . 3
- Launching Cloud Manager from the AWS Marketplace . . . 6
- Deploying Cloud Manager from the Azure Marketplace . . . 8
- Deploying Cloud Manager in an Azure US Government region . . . 11
- Installing Cloud Manager in an Azure Germany region . . . 15

# Additional deployment options

## Cloud Manager host requirements

If you install Cloud Manager on your own host, then you must verify support for your configuration, which includes operating system requirements, port requirements, and so on.

> You can install Cloud Manager on your own host in GCP, but not in your on-premises network. Cloud Manager must be installed in GCP in order to deploy Cloud Volumes ONTAP in GCP.

**A dedicated host is required**

Cloud Manager is not supported on a host that is shared with other applications. The host must be a dedicated host.

**Supported AWS EC2 instance types**

- t2.medium
- t3.medium (recommended)
- m4.large
- m5.xlarge
- m5.2xlarge
- m5.4xlarge
- m5.8xlarge

**Supported Azure VM sizes**

A2, D2 v2, or D2 v3 (based on availability)

**Supported GCP machine types**

A machine type with at least 2 vCPUs and 4 GB of memory.

**Supported operating systems**

- CentOS 7.2
- CentOS 7.3
- CentOS 7.4
- CentOS 7.5
- Red Hat Enterprise Linux 7.2
- Red Hat Enterprise Linux 7.3

- Red Hat Enterprise Linux 7.4
- Red Hat Enterprise Linux 7.5

  The Red Hat Enterprise Linux system must be registered with Red Hat Subscription Management. If it is not registered, the system cannot access repositories to update required 3rd party software during Cloud Manager installation.

  Cloud Manager is supported on English-language versions of these operating systems.

**Hypervisor**

A bare metal or hosted hypervisor that is certified to run CentOS or Red Hat Enterprise Linux
Red Hat Solution: Which hypervisors are certified to run Red Hat Enterprise Linux?

**CPU**

2.27 GHz or higher with two cores

**RAM**

4 GB

**Free disk space**

50 GB

**Outbound internet access**

Outbound internet access is required when installing Cloud Manager and when using Cloud Manager to deploy Cloud Volumes ONTAP. For a list of endpoints, see Networking requirements for Cloud Manager.

**Ports**

The following ports must be available:

- 80 for HTTP access
- 443 for HTTPS access
- 3306 for the Cloud Manager database
- 8080 for the Cloud Manager API proxy

  If other services are using these ports, Cloud Manager installation fails.

  > There is a potential conflict with port 3306. If another instance of MySQL is running on the host, it uses port 3306 by default. You must change the port that the existing MySQL instance uses.

  You can change the default HTTP and HTTPS ports when you install Cloud Manager. You cannot change the default port for the MySQL database. If you change the HTTP and HTTPS ports, you must ensure that users can access the Cloud Manager web console from a remote host:

- Modify the security group to allow inbound connections through the ports.
- Specify the port when you enter the URL to the Cloud Manager web console.

# Installing Cloud Manager on an existing Linux host

The most common way to deploy Cloud Manager is from Cloud Central or from a cloud provider's marketplace. But you have the option to download and install the Cloud Manager software on an existing Linux host in your network or in the cloud.

> You can install Cloud Manager on your own host in GCP, but not in your on-premises network. Cloud Manager must be installed in GCP in order to deploy Cloud Volumes ONTAP in GCP.

*Before you begin*

- A Red Hat Enterprise Linux system must be registered with Red Hat Subscription Management. If it is not registered, the system cannot access repositories to update required 3rd party software during Cloud Manager installation.
- The Cloud Manager installer accesses several URLs during the installation process. You must ensure that outbound internet access is allowed to those endpoints. Refer to Networking requirements for Cloud Manager.

*About this task*

- Root privileges are not required to install Cloud Manager.
- Cloud Manager installs the AWS command line tools (awscli) to enable recovery procedures from NetApp support.

  If you receive a message that installing the awscli failed, you can safely ignore the message. Cloud Manager can operate successfully without the tools.

- The installer that is available on the NetApp Support Site might be an earlier version. After installation, Cloud Manager automatically updates itself if a new version is available.

*Steps*

1. Review networking requirements:
   - Networking requirements for Cloud Manager
   - Networking requirements for Cloud Volumes ONTAP in AWS
   - Networking requirements for Cloud Volumes ONTAP in Azure
   - Networking requirements for Cloud Volumes ONTAP in GCP
2. Review Cloud Manager host requirements.
3. Download the software from the NetApp Support Site, and then copy it to the Linux host.

For help with connecting and copying the file to an EC2 instance in AWS, see AWS Documentation: Connecting to Your Linux Instance Using SSH.

4. Assign permissions to execute the script.

   **Example**

   ```
   chmod +x OnCommandCloudManager-V3.7.0.sh
   ```

5. Run the installation script:

   ```
   ./OnCommandCloudManager-V3.7.0.sh [silent] [proxy=ipaddress] [proxyport=port]
   [proxyuser=user_name] [proxypwd=password]
   ```

   *silent* runs the installation without prompting you for information.

   *proxy* is required if the Cloud Manager host is behind a proxy server.

   *proxyport* is the port for the proxy server.

   *proxyuser* is the user name for the proxy server, if basic authentication is required.

   *proxypwd* is the password for the user name that you specified.

6. Unless you specified the silent parameter, type **Y** to continue the script, and then enter the HTTP and HTTPS ports when prompted.

   If you change the HTTP and HTTPS ports, you must ensure that users can access the Cloud Manager web console from a remote host:

   - Modify the security group to allow inbound connections through the ports.
   - Specify the port when you enter the URL to the Cloud Manager web console.

     Cloud Manager is now installed. At the end of the installation, the Cloud Manager service (occm) restarts twice if you specified a proxy server.

7. Open a web browser and enter the following URL:

   https://*ipaddress*:*port*

   *ipaddress* can be localhost, a private IP address, or a public IP address, depending on the configuration of the Cloud Manager host. For example, if Cloud Manager is in the public cloud without a public IP address, you must enter a private IP address from a host that has a connection to the Cloud Manager host.

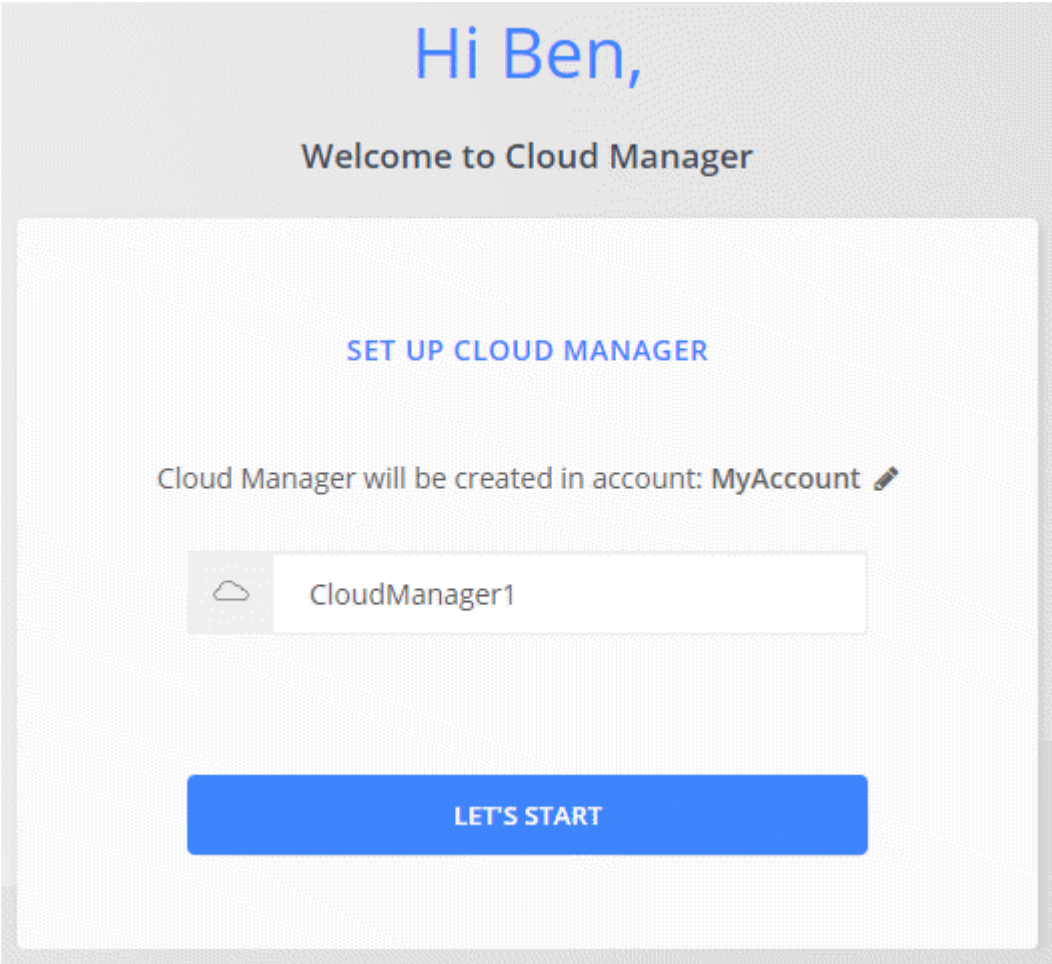
   *port* is required if you changed the default HTTP (80) or HTTPS (443) ports. For example, if the

HTTPS port was changed to 8443, you would enter https://*ipaddress*:8443

8. Sign up at NetApp Cloud Central or log in.
9. After you log in, set up Cloud Manager:
   a. Specify the Cloud Central account to associate with this Cloud Manager system.

      Learn about Cloud Central accounts.

   b. Enter a name for the system.




*After you finish*

Set up permissions so Cloud Manager can deploy Cloud Volumes ONTAP in your cloud provider:

- AWS: Set up an AWS account and then add it to Cloud Manager.
- Azure: Set up an Azure account and then add it to Cloud Manager.
- GCP: Set up a service account that has the permissions that Cloud Manager needs to create and manage Cloud Volumes ONTAP systems in projects.
  1. Create a role in GCP that includes the permissions defined in the Cloud Manager policy for GCP.
  2. Create a GCP service account and apply the custom role that you just created.

3. Associate this service account with the Cloud Manager VM.
4. If you want to deploy Cloud Volumes ONTAP in other projects, grant access by adding the service account with the Cloud Manager role to that project. You'll need to repeat this step for each project.

# Launching Cloud Manager from the AWS Marketplace

It's best to launch Cloud Manager in AWS using NetApp Cloud Central, but you can launch it from the AWS Marketplace, if needed.

> If you launch Cloud Manager from the AWS Marketplace, Cloud Manager is still integrated with NetApp Cloud Central. Learn more about the integration.

*About this task*

The following steps describe how to launch the instance from the EC2 Console because the console enables you to attach an IAM role to the Cloud Manager instance. This is not possible using the **Launch from Website** action.

*Steps*

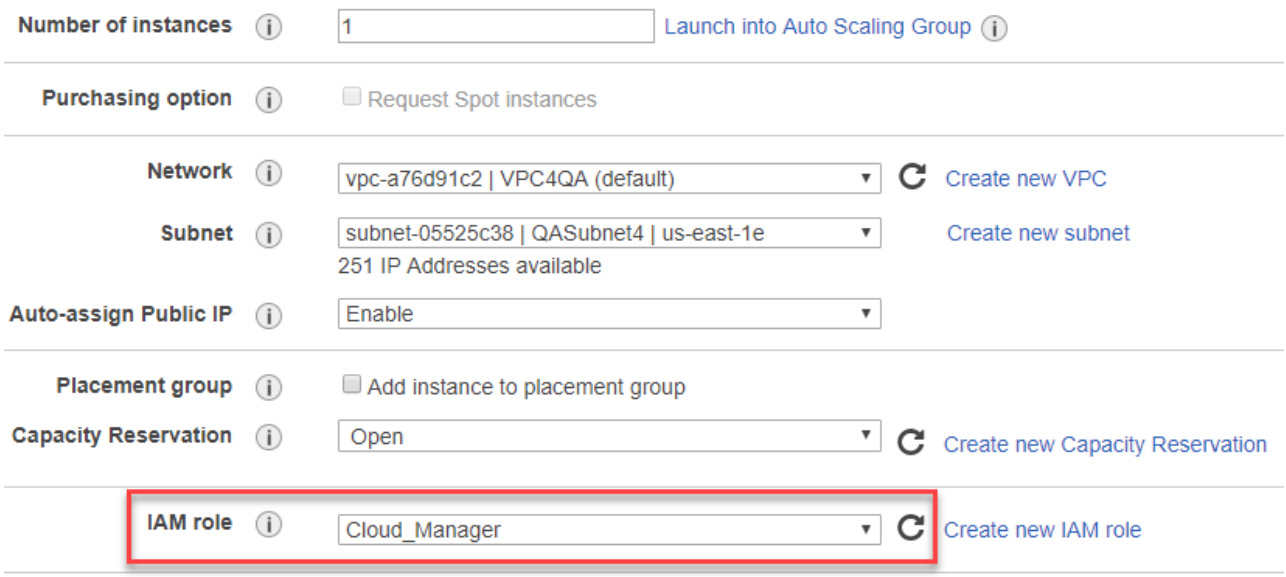
1. Create an IAM policy and role for the EC2 instance:
   a. Download the Cloud Manager IAM policy from the following location:

      NetApp Cloud Manager: AWS, Azure, and GCP Policies

   b. From the IAM console, create your own policy by copying and pasting the text from the Cloud Manager IAM policy.
   c. Create an IAM role with the role type Amazon EC2 and attach the policy that you created in the previous step to the role.
2. Subscribe from the AWS Marketplace to ensure that there's no disruption of service after your free trial of Cloud Volumes ONTAP ends. You'll be charged from this subscription for every Cloud Volumes ONTAP 9.6 and later PAYGO system that you create and each add-on feature that you enable.
3. Now go to the Cloud Manager page on the AWS Marketplace to deploy Cloud Manager from an AMI.
4. On the Marketplace page, click **Continue to Subscribe** and then click **Continue to Configuration**.
5. Change any of the default options and click **Continue to Launch**.
6. Under **Choose Action**, select **Launch through EC2** and then click **Launch**.
7. Follow the prompts to configure and deploy the instance:
   - **Choose Instance Type**: Depending on region availability, choose one of the supported instance types (t3.medium is recommended).

Review the list of supported instance types.

- **Configure Instance**: Select a VPC and subnet, the IAM role that you created in step 1, and other configuration options that meet your requirements.




- **Add Storage**: Keep the default storage options.
- **Add Tags**: Enter tags for the instance, if desired.
- **Configure Security Group**: Specify the required connection methods for the Cloud Manager instance: SSH, HTTP, and HTTPS.
- **Review**: Review your selections and click **Launch**.

  AWS launches the software with the specified settings. The Cloud Manager instance and software should be running in approximately five minutes.

8. Open a web browser from a host that has a connection to the Cloud Manager virtual machine and enter the following URL:

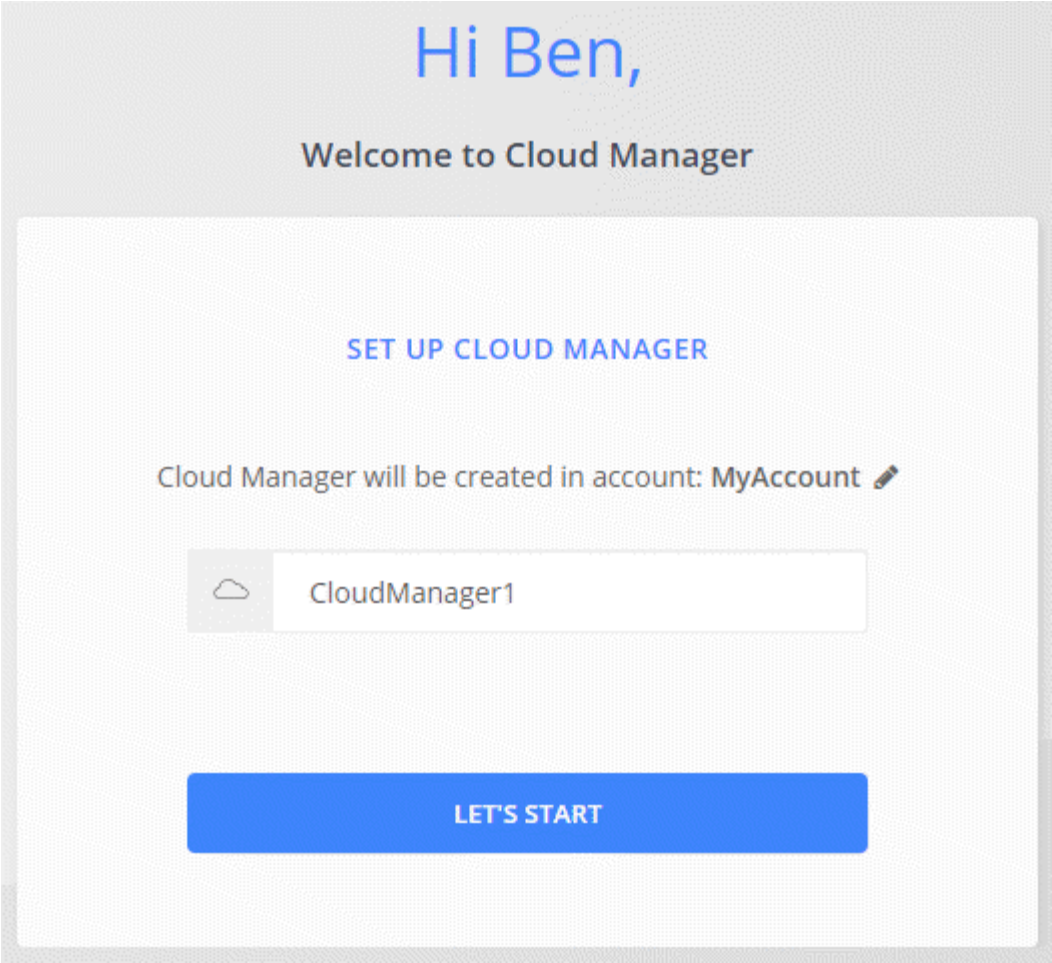
   http://*ipaddress*:80

9. After you log in, set up Cloud Manager:

   a. Specify the Cloud Central account to associate with this Cloud Manager system.

      Learn about Cloud Central accounts.

   b. Enter a name for the system.




*Result*

Cloud Manager is now installed and set up.

# Deploying Cloud Manager from the Azure Marketplace

It is best to deploy Cloud Manager in Azure using NetApp Cloud Central, but you can deploy it from the Azure Marketplace, if needed.

Separate instructions are available to deploy Cloud Manager in Azure US Government regions and in Azure Germany regions.

> If you deploy Cloud Manager from the Azure Marketplace, Cloud Manager is still integrated with NetApp Cloud Central. Learn more about the integration.

## Deploying Cloud Manager in Azure

You need to install and set up Cloud Manager so you can use it to launch Cloud Volumes ONTAP in Azure.

*Steps*

1. Go to the Azure Marketplace page for Cloud Manager.
2. Click **Get it now** and then click **Continue**.
3. From the Azure portal, click **Create** and follow the steps to configure the virtual machine.

   Note the following as you configure the VM:

   - Cloud Manager can perform optimally with either HDD or SSD disks.
   - Choose one of the recommended virtual machine sizes: A2, D2 v2, or D2 v3 (based on availability).
   - For the network security group, Cloud Manager requires inbound connections using SSH, HTTP, and HTTPS.

     Learn more about security group rules for Cloud Manager.

   - Under **Management**, enable **System assigned managed identity** for Cloud Manager by selecting **On**.

     This setting is important because a managed identity allows the Cloud Manager virtual machine to identify itself to Azure Active Directory without providing any credentials. Learn more about managed identities for Azure resources.

4. On the **Review + create** page, review your selections and click **Create** to start the deployment.

   Azure deploys the virtual machine with the specified settings. The virtual machine and Cloud Manager software should be running in approximately five minutes.

5. Open a web browser from a host that has a connection to the Cloud Manager virtual machine and enter the following URL:

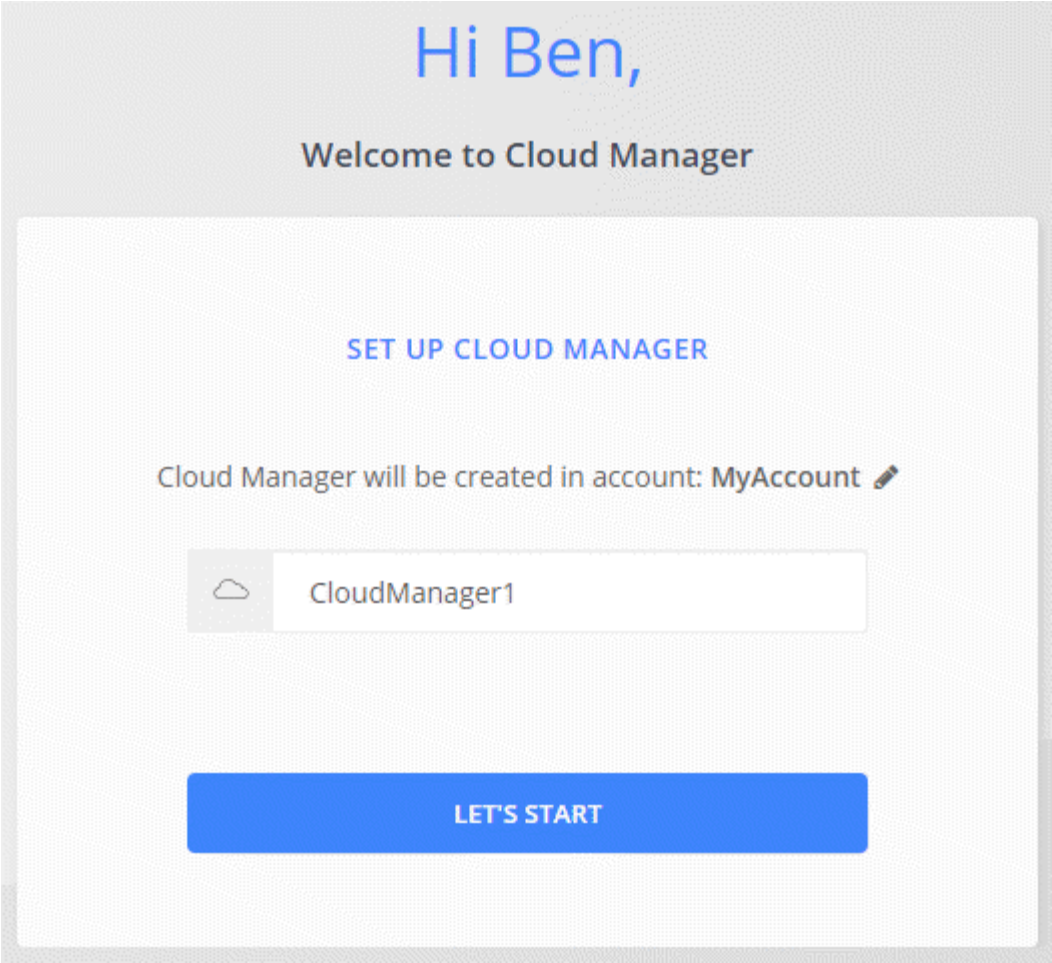
   http://*ipaddress*:80

6. After you log in, set up Cloud Manager:

   a. Specify the Cloud Central account to associate with this Cloud Manager system.

      Learn about Cloud Central accounts.

   b. Enter a name for the system.




*Result*

Cloud Manager is now installed and set up. You must grant Azure permissions before users can deploy Cloud Volumes ONTAP in Azure.

## Granting Azure permissions to Cloud Manager

When you deployed Cloud Manager in Azure, you should have enabled a system-assigned managed identity. You must now grant the required Azure permissions by creating a custom role and then by assigning the role to the Cloud Manager virtual machine for one or more subscriptions.

*Steps*

1. Create a custom role using the Cloud Manager policy:
   a. Download the Cloud Manager Azure policy.
   b. Modify the JSON file by adding Azure subscription IDs to the assignable scope.

      You should add the ID for each Azure subscription from which users will create Cloud Volumes ONTAP systems.

      **Example**

```
"AssignableScopes": [
"/subscriptions/d333af45-0d07-4154-943d-c25fbzzzzzzz",
"/subscriptions/54b91999-b3e6-4599-908e-416e0zzzzzzz",
"/subscriptions/398e471c-3b42-4ae7-9b59-ce5bbzzzzzzz"
```

c. Use the JSON file to create a custom role in Azure.

   The following example shows how to create a custom role using the Azure CLI 2.0:

   **az role definition create --role-definition C:\Policy_for_cloud_Manager_Azure_3.7.4.json**

   You should now have a custom role called OnCommand Cloud Manager Operator that you can assign to the Cloud Manager virtual machine.

2. Assign the role to the Cloud Manager virtual machine for one or more subscriptions:

   a. Open the **Subscriptions** service and then select the subscription in which you want to deploy Cloud Volumes ONTAP systems.

   b. Click **Access control (IAM)**.

   c. Click **Add** > **Add role assignment** and then add the permissions:

      - Select the **OnCommand Cloud Manager Operator** role.

      > OnCommand Cloud Manager Operator is the default name provided in the Cloud Manager policy. If you chose a different name for the role, then select that name instead.

      - Assign access to a **Virtual Machine**.
      - Select the subscription in which the Cloud Manager virtual machine was created.
      - Select the Cloud Manager virtual machine.
      - Click **Save**.

   d. If you want to deploy Cloud Volumes ONTAP from additional subscriptions, switch to that subscription and then repeat these steps.

*Result*

Cloud Manager now has the permissions that it needs to deploy and manage Cloud Volumes ONTAP in Azure.

# Deploying Cloud Manager in an Azure US Government region

To get Cloud Manager up and running in a US Government region, first deploy Cloud Manager from the Azure Government Marketplace. Then provide the

permissions that Cloud Manager needs to deploy and manage Cloud Volumes ONTAP systems.

For a list of supported Azure US Government regions, see Cloud Volumes Global Regions.

## Deploying Cloud Manager from the Azure US Government Marketplace

Cloud Manager is available as an image in the Azure US Government Marketplace.

*Steps*

1. Ensure that the Azure Government Marketplace is enabled in your subscription:
   a. Log into the portal as an Enterprise Administrator.
   b. Navigate to **Manage**.
   c. Under **Enrollment Details**, click the pencil icon next to **Azure Marketplace**.
   d. Select **Enabled**.
   e. Click **Save**.

   Microsoft Azure Documentation: Azure Government Marketplace

2. Search for OnCommand Cloud Manager in the Azure US Government portal.
3. Click **Create** and follow the steps to configure the virtual machine.

   Note the following as you configure the virtual machine:

   - Cloud Manager can perform optimally with either HDD or SSD disks.
   - You should choose one of the recommended virtual machine sizes: A2, D2 v2, or D2 v3 (based on availability).
   - For the network security group, it is best to choose **Advanced**.

     The **Advanced** option creates a new security group that includes the required inbound rules for Cloud Manager. If you choose Basic, refer to Security group rules for the list of required rules.

4. On the summary page, review your selections and click **Create** to start the deployment.

   Azure deploys the virtual machine with the specified settings. The virtual machine and Cloud Manager software should be running in approximately five minutes.

5. Open a web browser from a host that has a connection to the Cloud Manager virtual machine and enter the following URL:

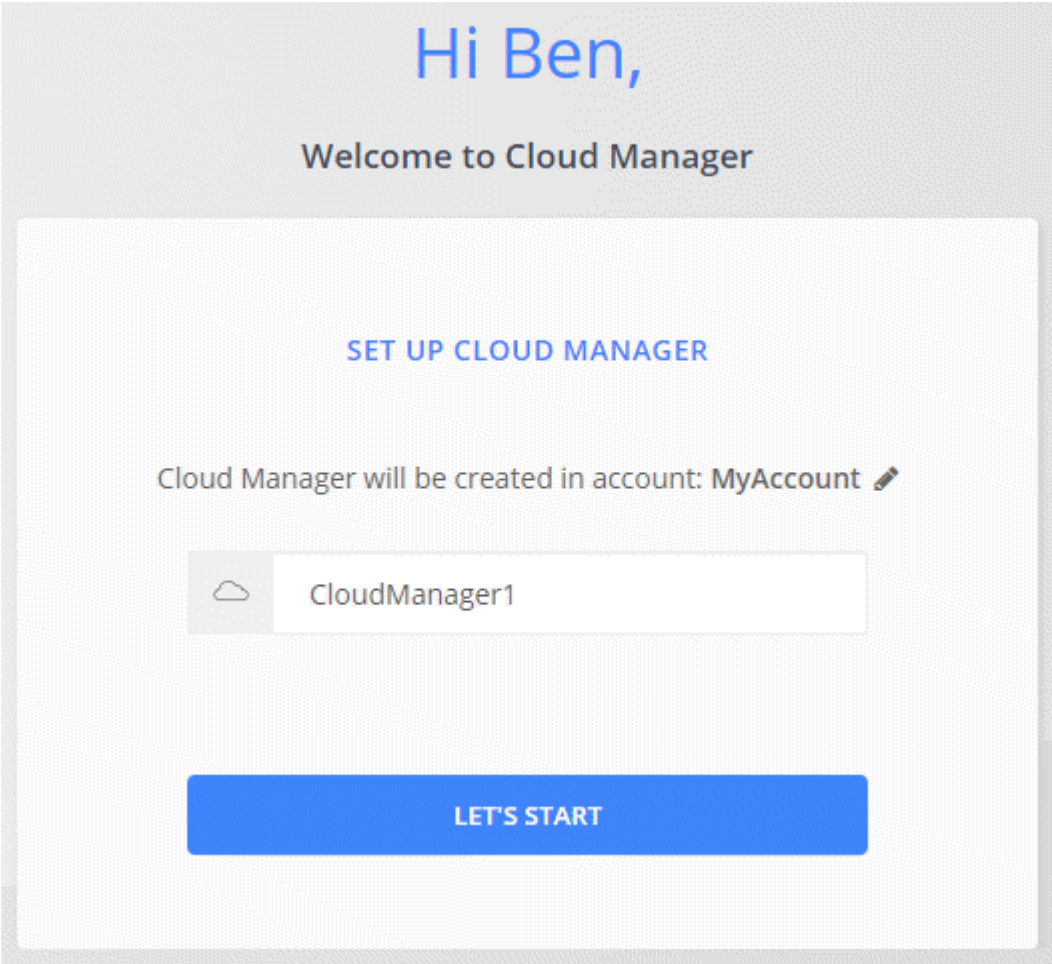
   http://*ipaddress*:80

6. After you log in, set up Cloud Manager:

a. Specify the Cloud Central account to associate with this Cloud Manager system.

   Learn about Cloud Central accounts.

b. Enter a name for the system.




*Result*

Cloud Manager is now installed and set up. You must grant Azure permissions before users can deploy Cloud Volumes ONTAP in Azure.

## Granting Azure permissions to Cloud Manager using a managed identity

The easiest way to provide permissions is by enabling a managed identity on the Cloud Manager virtual machine and then by assigning the required permissions to the virtual machine. If preferred, an alternative way is to grant Azure permissions using a service principal.

*Steps*

1. Enable a managed identity on the Cloud Manager virtual machine:

   a. Navigate to the Cloud Manager virtual machine and select **Identity**.

   b. Under **System Assigned**, click **On** and then click **Save**.

2. Create a custom role using the Cloud Manager policy:
   a. Download the Cloud Manager Azure policy.
   b. Modify the JSON file by adding Azure subscription IDs to the assignable scope.

      You should add the ID for each Azure subscription from which users will create Cloud Volumes ONTAP systems.

      **Example**

      "AssignableScopes": [
      "/subscriptions/d333af45-0d07-4154-943d-c25fbzzzzzzz",
      "/subscriptions/54b91999-b3e6-4599-908e-416e0zzzzzzz",
      "/subscriptions/398e471c-3b42-4ae7-9b59-ce5bbzzzzzzz"

   c. Use the JSON file to create a custom role in Azure.

      The following example shows how to create a custom role using the Azure CLI 2.0:

      **az role definition create --role-definition C:\Policy_for_cloud_Manager_Azure_3.7.4.json**

      You should now have a custom role called OnCommand Cloud Manager Operator that you can assign to the Cloud Manager virtual machine.

3. Assign the role to the Cloud Manager virtual machine for one or more subscriptions:
   a. Open the **Subscriptions** service and then select the subscription in which you want to deploy Cloud Volumes ONTAP systems.
   b. Click **Access control (IAM)**.
   c. Click **Add**, click **Add role assignment**, and then add the permissions:
      - Select the **OnCommand Cloud Manager Operator** role.

        OnCommand Cloud Manager Operator is the default name provided in the Cloud Manager policy. If you chose a different name for the role, then select that name instead.

      - Assign access to a **Virtual Machine**.
      - Select the subscription in which the Cloud Manager virtual machine was created.
      - Type the name of the virtual machine and then select it.
      - Click **Save**.
   d. If you want to deploy Cloud Volumes ONTAP from additional subscriptions, switch to that subscription and then repeat these steps.

*Result*

Cloud Manager now has the permissions that it needs to deploy and manage Cloud Volumes ONTAP in

Azure.

# Installing Cloud Manager in an Azure Germany region

The Azure Marketplace is not available in the Azure Germany regions, so you must download the Cloud Manager installer from the NetApp Support Site and install it on an existing Linux host in the region.

*Steps*

1. Review networking requirements for Azure.
2. Review Cloud Manager host requirements.
3. Download and install Cloud Manager.
4. Grant Azure permissions to Cloud Manager using a service principal.

*After you finish*

Cloud Manager is now ready to deploy Cloud Volumes ONTAP in the Azure Germany region, just like any other region. However, you might want to perform additional setup first.

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